

PYTHON – ASSIGNMENT 1

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DATA STRUCTURES – STRINGS & TUPLES

1. Project Title

Python Data Structures – Strings and Tuples

2. Introduction

Python provides several built-in data structures and features for storing and manipulating data. This assignment focuses on two important concepts: **Strings and Tuples**.

Strings are sequences of characters and can be manipulated using concatenation, indexing, slicing, and built-in string methods. Tuples are ordered collections of elements that can be accessed using indexing and slicing.

This assignment provides practical experience in working with strings and tuples using Python programming.

3. Objective

The main objectives of this assignment are:

- To understand the basic concept of Python strings.
- To perform string concatenation.
- To use user input in a Python program.
- To understand string indexing.
- To perform string slicing.
- To reverse a string using slicing.
- To extract specific words from a string.
- To use built-in string methods.

- To count occurrences of a character.
 - To replace a substring with another string.
 - To create and manipulate tuples.
 - To concatenate two tuples.
 - To repeat tuple elements.
 - To access individual tuple elements.
 - To access specific portions of a tuple using slicing.
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5. Assignment Tasks

The assignment is divided into the following sections:

1. String Concatenation
2. String Slicing and Indexing
3. String Methods
4. Tuples – Creation, Modification and Access

The original assignment specifically requires these operations.

6. Task 1 – String Concatenation

6.1 Objective

The objective of this task is to concatenate two strings. The first string is "Hello " and the second string is obtained from the user as input.

After displaying the greeting, another string ", welcome to Python programming" is concatenated with the existing string.

6.2 Program

```
string1 = "Hello "
```

```
name = input("Enter your Name: ")
```

```
result = string1 + name
```

```
print(result)
```

```
string3 = ", welcome to Python programming"
```

```
result = result + string3
```

```
print(result)
```

7. Task 2 – String Slicing and Indexing

7.1 Objective

The objective of this task is to perform indexing and slicing operations on the concatenated string.

The assignment requires the following operations:

- Print the first character.
 - Print the last character.
 - Print the first five characters.
 - Print the last eleven characters.
 - Reverse the string.
 - Extract the word "Python".
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8. Task 3 – String Methods

8.1 Objective

The objective of this task is to use Python's built-in string methods on:

```
strM = "Python beginner tutorial"
```

The assignment requires conversion to uppercase and lowercase, capitalization, counting the character t, and replacing "Python" with "Machine Learning".

8.2 Program

```
strM = "Python beginner tutorial"
```

```
print("Uppercase:")
```

```
print (strM.upper())
```

```
print("Lowercase:")
```

```
print (strM.lower())
```

```
print ("Capitalize:")
```

```
print (strM.capitalize())
```

```
print ("Number of 't' characters:")
```

```
print (strM.count('t'))
```

```
print ("After replacement:")
```

```
print (strM.replace("Python", "Machine Learning"))
```

9. Task 4 – Tuples

9.1 Objective

The objective of this task is to create and manipulate tuples.

The assignment requires two tuples:

(10, 20, 30)

and

(40, 50, 60)

The two tuples must then be combined, repeated three times, and accessed using indexing and slicing.

9.2 Creating the Tuples

Program

```
t1 = (10, 20, 30)
```

```
t2 = (40, 50, 60)
```

```
print(t1)
```

```
print(t2)
```

Output

```
(10, 20, 30)
```

```
(40, 50, 60)
```

9.3 Concatenating the Tuples

The two tuples are combined using the + operator.

Program

```
t_combine = t1 + t2
```

```
print(t_combine)
```

Output

```
(10, 20, 30, 40, 50, 60)
```

Explanation

The + operator combines the elements of the two tuples into a single tuple.

9.4 Repeating the Tuple Three Times

Program

```
t_repeat = t_combine * 3
```

```
print(t_repeat)
```

Output

(10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)

Explanation

The * operator is used to repeat the tuple three times.

9.5 Accessing the Third Element

Program

```
print(t_combine[2])
```

Output

30

Explanation

Python uses zero-based indexing:

Index 0 → 10

Index 1 → 20

Index 2 → 30

Therefore, index 2 accesses the third element.

9.6 Accessing the First Three Elements

Program

```
print(t_combine[:3])
```

Output

(10, 20, 30)

Explanation

The slicing operation [:3] returns the first three elements.

9.7 Accessing the Last Three Elements

Program

```
print(t_combine[-3:])
```

Output

(40, 50, 60)

Explanation

The slicing operation [-3:] returns the last three elements.

10. Concepts Learned

Through this assignment, the following Python concepts were practiced:

Strings

- String declaration
- String concatenation
- User input
- String indexing
- Positive indexing
- Negative indexing
- String slicing
- Reverse slicing
- String methods
- upper()
- lower()
- capitalize()
- count()
- replace()

Tuples

- Tuple creation

- Tuple concatenation
 - Tuple repetition
 - Tuple indexing
 - Tuple slicing
 - Accessing first and last elements
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11. Conclusion

The Python Fundamentals Assignment 1 was successfully completed using Python. The assignment provided practical understanding of **strings and tuples**, including concatenation, indexing, slicing, built-in string methods, tuple concatenation, tuple repetition, and accessing tuple elements.

12. Final Result

Status: Assignment Completed Successfully

The following topics were successfully implemented:

- String concatenation
- String indexing
- String slicing
- String reversal
- Word extraction
- Uppercase conversion
- Lowercase conversion
- Capitalization
- Character counting
- String replacement
- Tuple creation
- Tuple concatenation

- Tuple repetition
- Tuple element access
- Tuple slicing